EOSC 454 Assignment 3

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1 Problem 1: Forward Problem

1.1 1a

Kernel functions of the form:

$$g_j(x) = e^{jpx} cos(2\pi jqx) \tag{1}$$

can be modified by changing any of the parameters p, q or j. See figure 1.

p

Modifying p changes the rate of growth or decay of the kernel. Negative p values result in kernel decay, while positive result in kernel growth, assuming $j > 0$.

q

Modifying q changes how oscillatory the kernel is. Higher values of q result in more oscillatory kernels.

j

Modifying j changes both how oscillatory a kernel is and how fast it decays. Assuming p is negative, higher j values result in a faster decaying, more oscillatory function

1.2 1b

See figure 2.

1.3 1c

See figure 3.

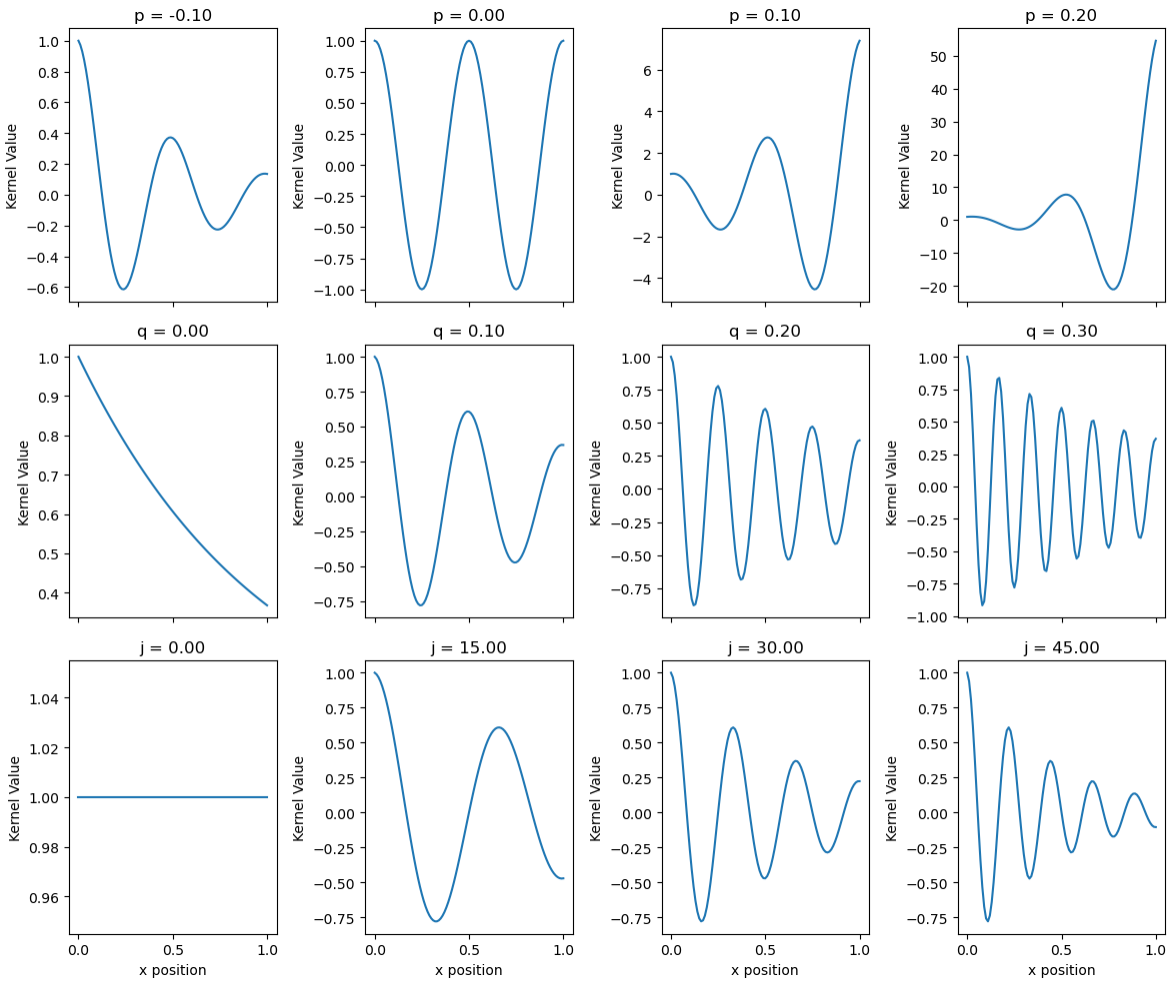

Figure 1: Figure showing the variation of the parameters p, q, j. In each row of the plot, one parameter is varied while the others are held fixed.

1.4 1d

Using the formula

$$s = \frac{\|\mathbf{d}(\mathbf{m_1}) - \mathbf{d}(\mathbf{m_2})\|}{\|\Delta m\|} \tag{2}$$

to compute sensitivity of the data to changes in the boxcar and gaussian amplitude, we find that for this model, the data is significantly more sensitve to gaussian amplitude. This may be due to the fact that the gaussian takes up more space in the model, causing a change in the amplitude to affect the data more.

Table 1: Sensitivity Analysis of Observed Data

Model Feature	**Sensitivity Value**
Boxcar Amplitude	0.3237
Gaussian Amplitude	0.5039

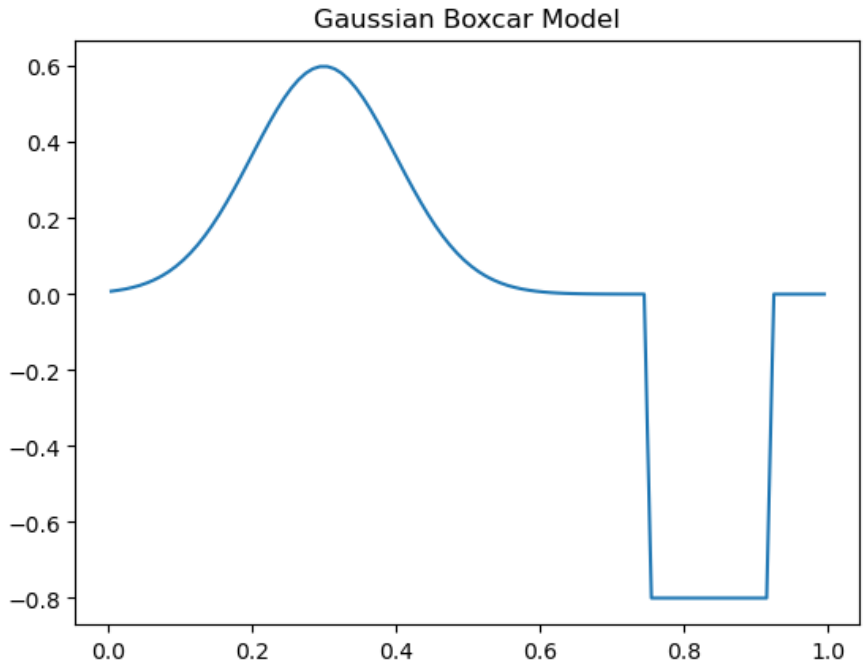

Figure 2: The model used in this assignment, consisting of a positive amplitude gaussian distribution and a negative amplitude boxcar.

1.5 1e

Once again using equation 2 to quantify sensitivity, this time we vary q to see how changing the frequency of the kernels are affects the sensitivity, as shown in table 2.

Table 2: Sensitivity Comparison Across Spatial Frequencies (q)

Model Feature	**Sensitivity** ($q = 0.05$)	**Sensitivity** ($q = 0.2$)
Boxcar Amplitude	0.3394	0.2892
Gaussian Amplitude	0.5970	0.3948

As before, the sensitivity to changes in gaussian amplitude is higher than sensitivity to changes in boxcar amplitude. We also observe that sensitivity decreases for higher frequency kernels, but the drop is much more significant for the gaussian. This may be due to the fact that the gaussian is smooth and consists of mostly low frequency components, while the boxcar has slightly higher frequency components due to its sharp edges.

1.6 1f

In this problem, we observe how changing the decay of the kernels affects sensitivity, as seen in table 3.

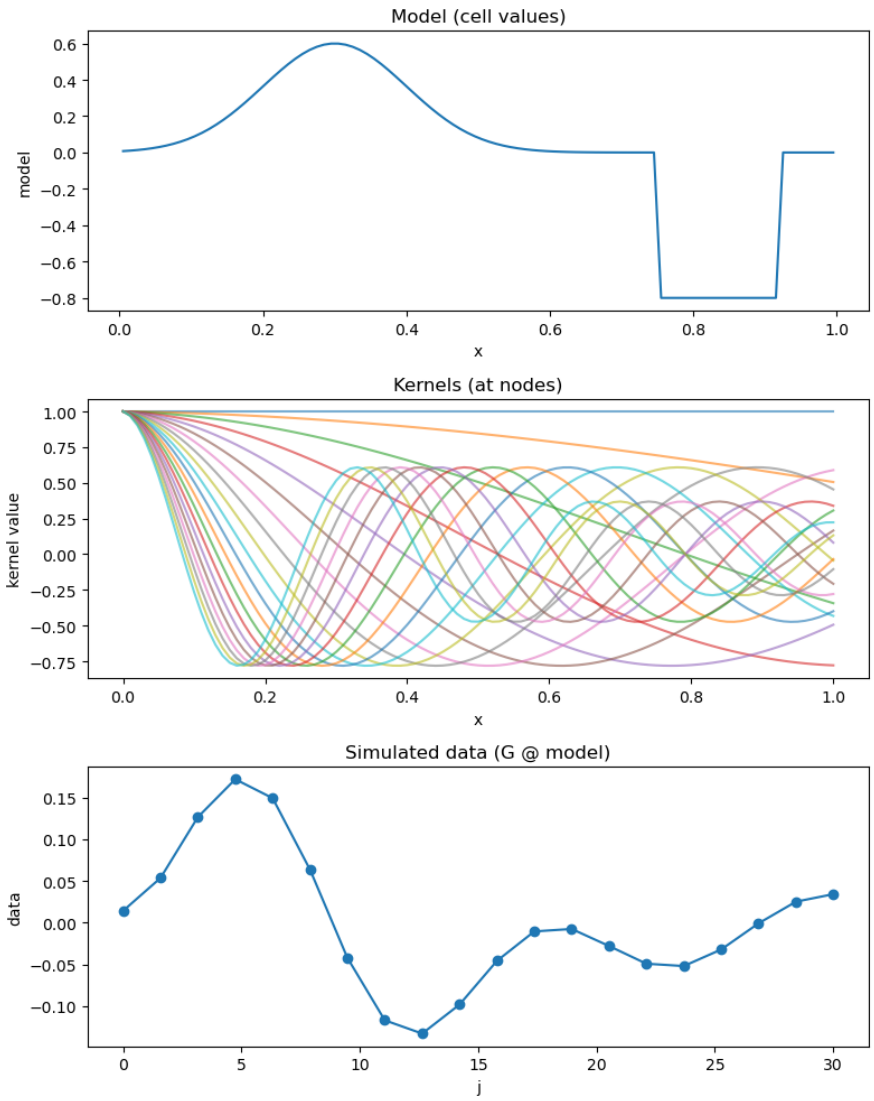

Figure 3: Plot showing the model, the kernels used, and the simulated data generated by performing the multiplication d = Gm.

Table 3: Sensitivity Analysis for Varying Decay Constants (p)

Model Feature	**Rapid Decay** ($p = -0.1$)	**Slow Decay** ($p = -0.01$)
Boxcar Amplitude	0.2609	0.4346
Gaussian Amplitude	0.4664	0.5440

In this section, we observe that sensitivity is higher for lower decay rates, with the jump in sensitivity being much larger for changes in the boxcar. This makes sense as the boxcar occurs near the end of the kernel, so increasing kernel amplitude in the region of the boxcar makes the model more sensitive to changes in that region.

2 Problem 2: Inverse Problem

2.1 2a

W_d has shape n_kernels x n_kernels, with values on the main diagonal. W_m has shape n_cells x n_cells, with values on the main diagonal. W_x has shape n_cells-1 x n_cells, with values on the main diagonal and one on the superdiagonal. See figure 4

```
Wd matrix:
[[100.   0.]
 [  0. 100.]]
Wm matrix:
[[0.5 0.  0.  0. ]
 [0.  0.5 0.  0. ]
 [0.  0.  0.5 0. ]
 [0.  0.  0.  0.5]]
Wx matrix:
[[-2.  2.  0.  0.]
 [ 0. -2.  2.  0.]
 [ 0.  0. -2.  0.]]
```

Figure 4: Sample matricies generated with n_kernels = 2 and n_cells = 4

2.2 2b

Want to find when $\nabla\phi = 0$.

$$\phi(\mathbf{m}) = \phi_{\mathbf{d}}(\mathbf{m}) + \beta\phi_{\mathbf{m}}(\mathbf{m}) \tag{3}$$

Take the gradient of each part individually:

$$\nabla_m \phi_d = \nabla_m \|\mathbf{W_d}(\mathbf{Gm} - \mathbf{d})\|^{\mathbf{2}}$$

Following the derivation covered in class we find that

$$0 = \mathbf{G}^\top \mathbf{W_d}^\top \mathbf{W_d} \mathbf{Gm} - \mathbf{G}^\top \mathbf{W_d}^\top \mathbf{W_d} \mathbf{d^{obs}}$$

$$\nabla_m \beta \phi_m = \nabla_m \beta(\alpha_s \|\mathbf{W}_s(\mathbf{m} - \mathbf{m}_{ref})\|^2 + \alpha_x \|\mathbf{W}_x \mathbf{m}\|^2)$$

Assuming $\mathbf{m_{ref}} = \mathbf{0}$ this becomes:

$$\beta(\alpha_s \mathbf{W_s}^\top \mathbf{W_s} \mathbf{m} + \alpha_\mathbf{x} \mathbf{W_x}^\top \mathbf{W_x} \mathbf{m}) = \mathbf{0}$$

Finally combining the two terms:

$$(\mathbf{G}^\top \mathbf{W_d}^\top \mathbf{W_d} \mathbf{G} + \beta(\alpha_\mathbf{s} \mathbf{W_s}^\top \mathbf{W_s} + \alpha_\mathbf{x} \mathbf{W_x}^\top \mathbf{W_x}))\mathbf{m} = \mathbf{G}^\top \mathbf{W_d}^\top \mathbf{W_d} \mathbf{d^{obs}} \tag{4}$$

2.3 2c

Scenario A: $\alpha_s = 1$, $\alpha_x = 0$. In this case, the ideal β value was approximately 1000, resulting in a ϕ_d value of 29.93. See figure 5.

Scenario B: $\alpha_s = 0$, $\alpha_x = 1$. In this case, the ideal β value was approximately 10, resulting in a ϕ_d value of 31.92. See figure 6.

3 Problem 3: Tikhonov Curves

3.1 3a

For model fits $\alpha_s = 1$, $\alpha_x = 0$, see figures 7, 8, 9.

3.2 3b

Repeating the previous section, now with $\alpha_s = 0$, $\alpha_x = 1$. See figures 10, 11, 12. Overall, these fits look fairly similar, with the largest difference being that the $\alpha_x = 1$ fits being less oscillatory. This means that in the underfit case, the smooth gaussian is fit much better for when $\alpha_x = 1$, but the sharp boxcar fit is worse. The overfit case is also much less oscillatory when $\alpha_x = 1$.

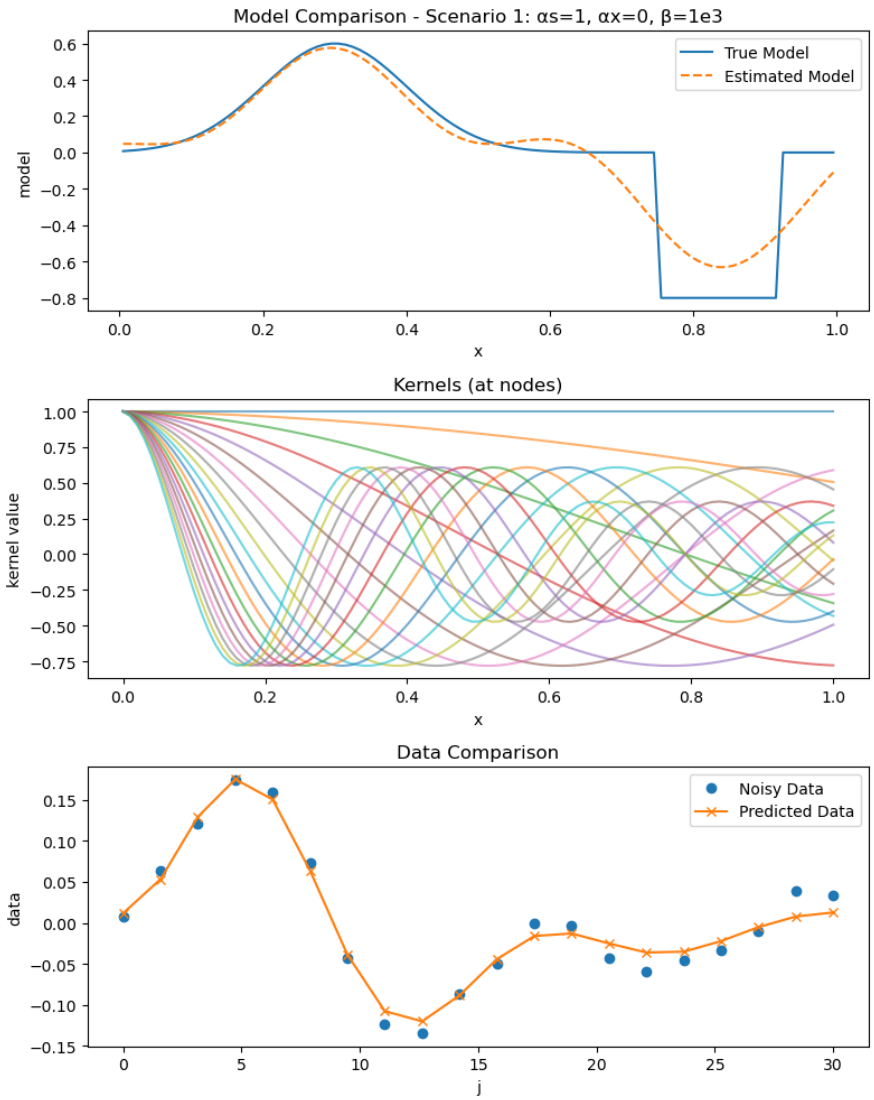

Figure 5: Model fit, kernels, data comparison with ideal beta using data with gaussian noise added. $\alpha_s = 1$, $\alpha_x = 0$

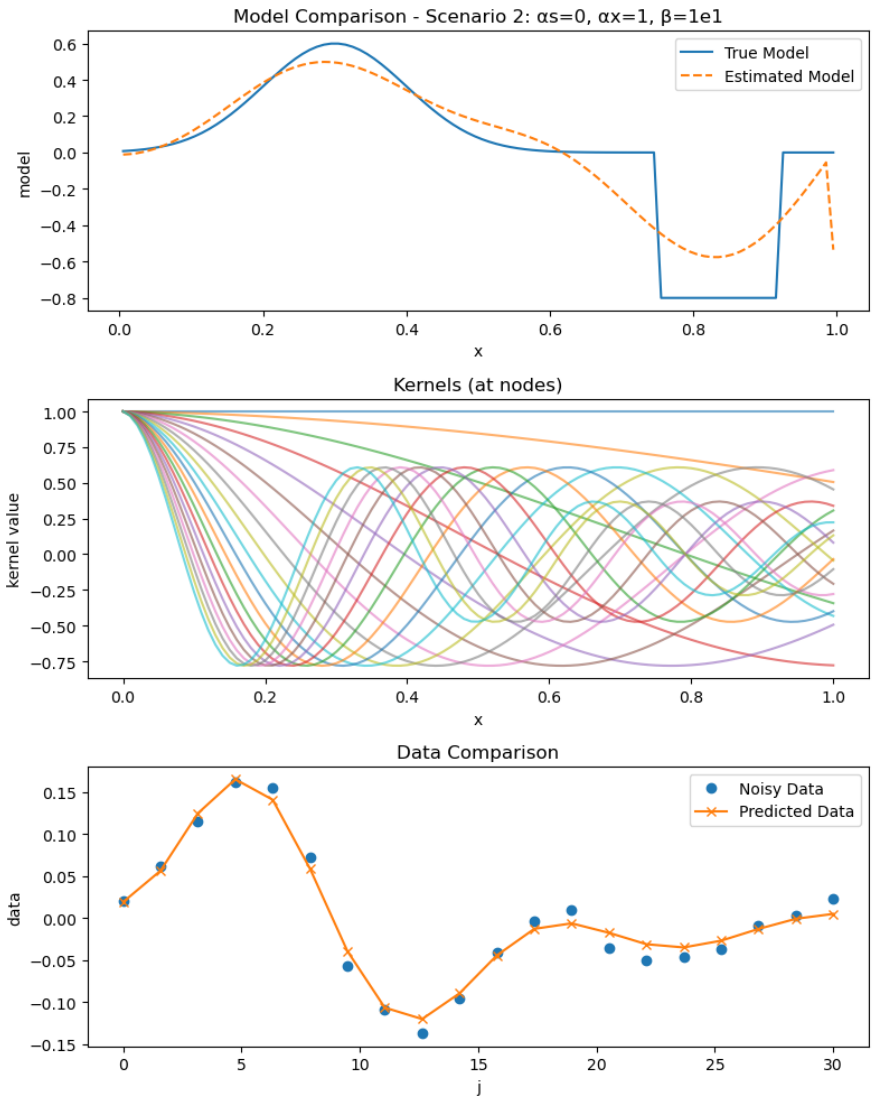

Figure 6: Model fit, kernels, data comparison with ideal beta using data with gaussian noise added. $\alpha_s = 0$, $\alpha_x = 1$

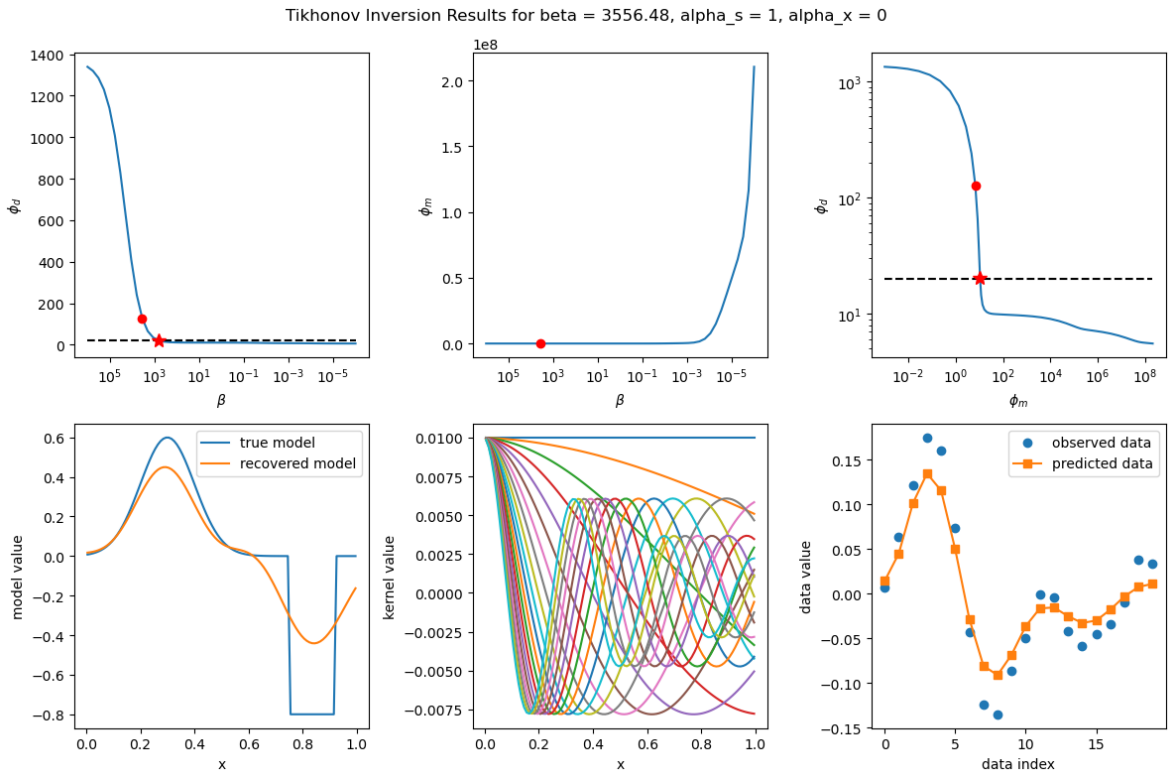

Figure 7: Tikhonov Curves, model, kernels, data for $\alpha_s = 1$, $\alpha_x = 0$. In this case we have not reached the target misfit and the data is underfit.

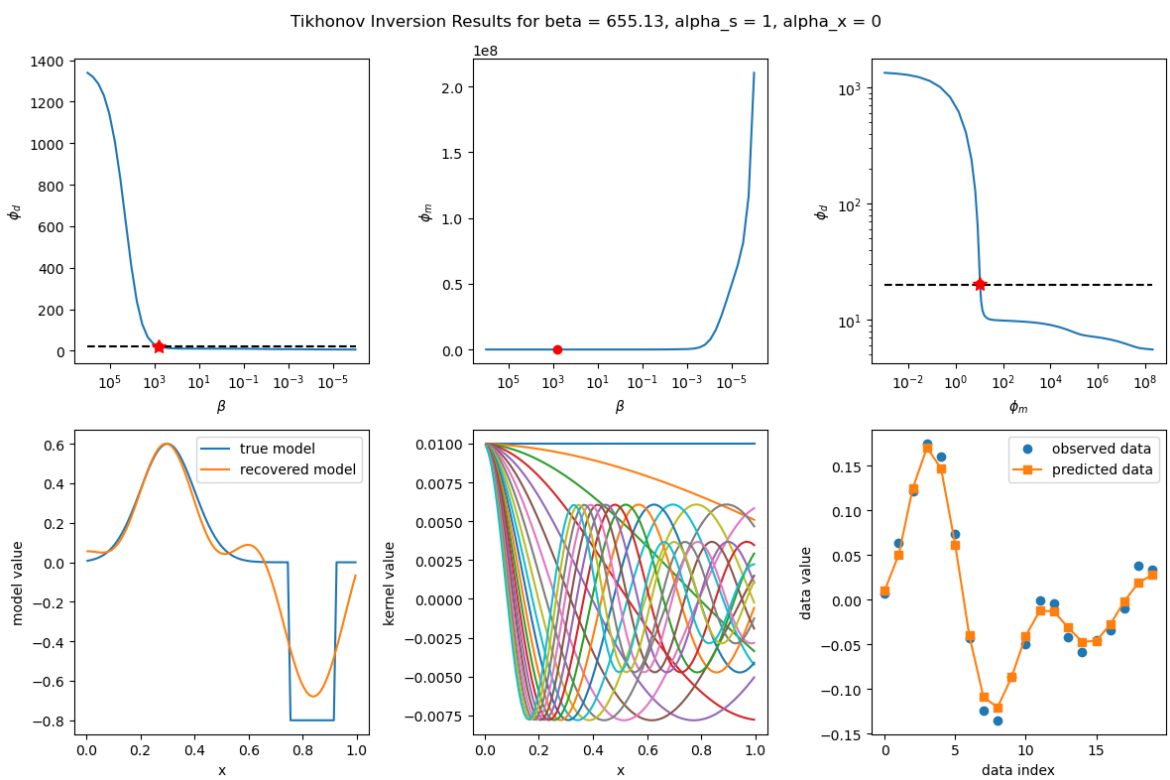

Figure 8: Tikhonov Curves, model, kernels, data for $\alpha_s = 1$, $\alpha_x = 0$. In this case we have reached the target misfit and the data fit is acceptable.

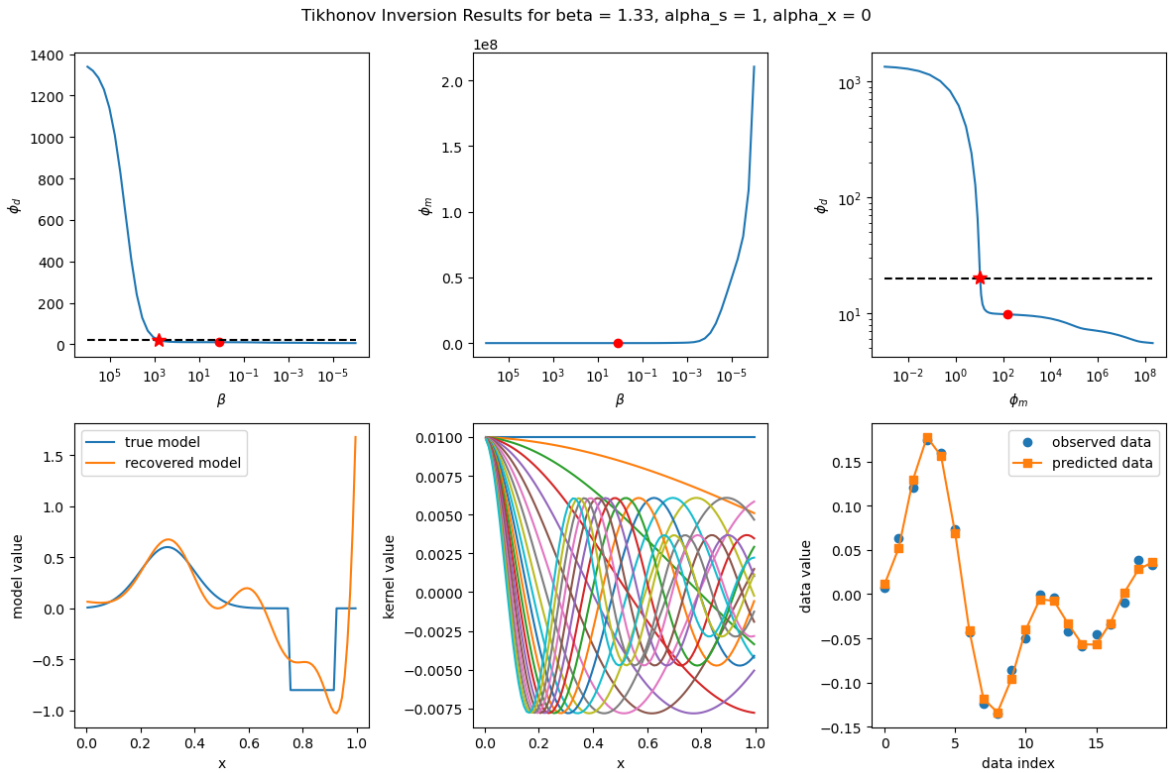

Figure 9: Tikhonov Curves, model, kernels, data for $\alpha_s = 1$, $\alpha_x = 0$. In this case we have overshot target misfit and the data is overfit, resulting in an oscillatory model.

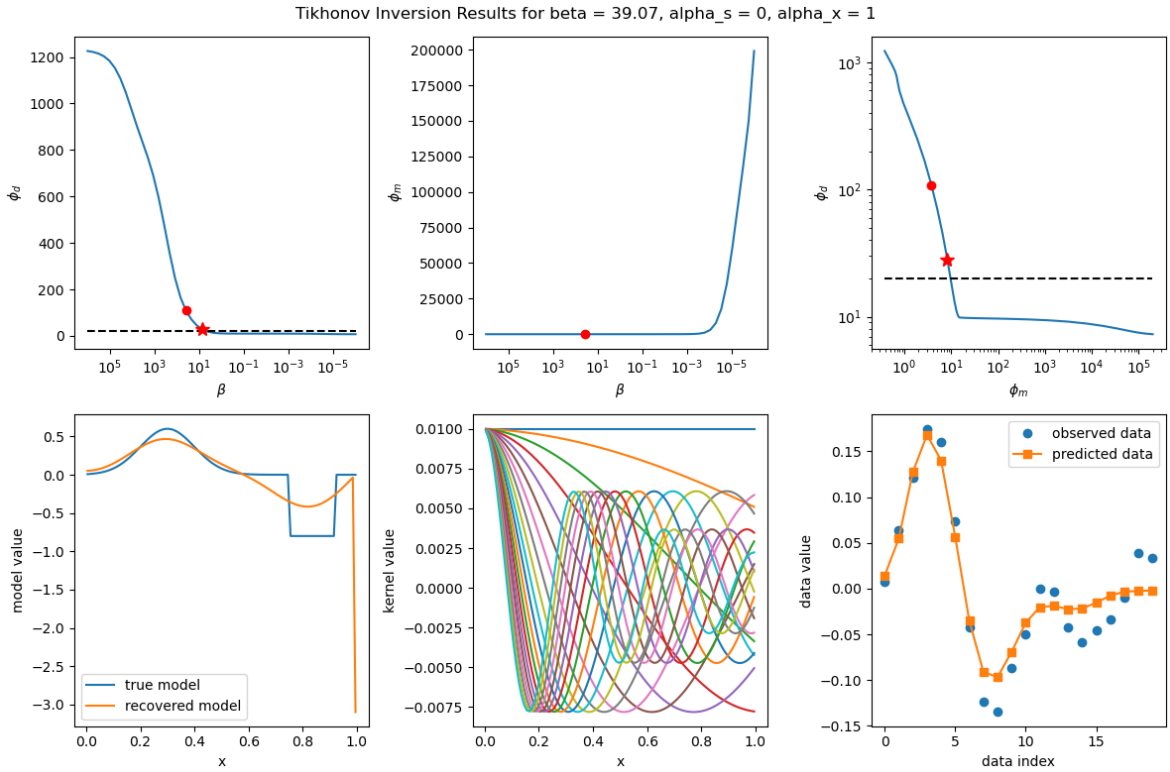

Figure 10: Tikhonov Curves, model, kernels, data for $\alpha_s = 0$, $\alpha_x = 1$. In this case we have not reached the target misfit and the data is underfit.

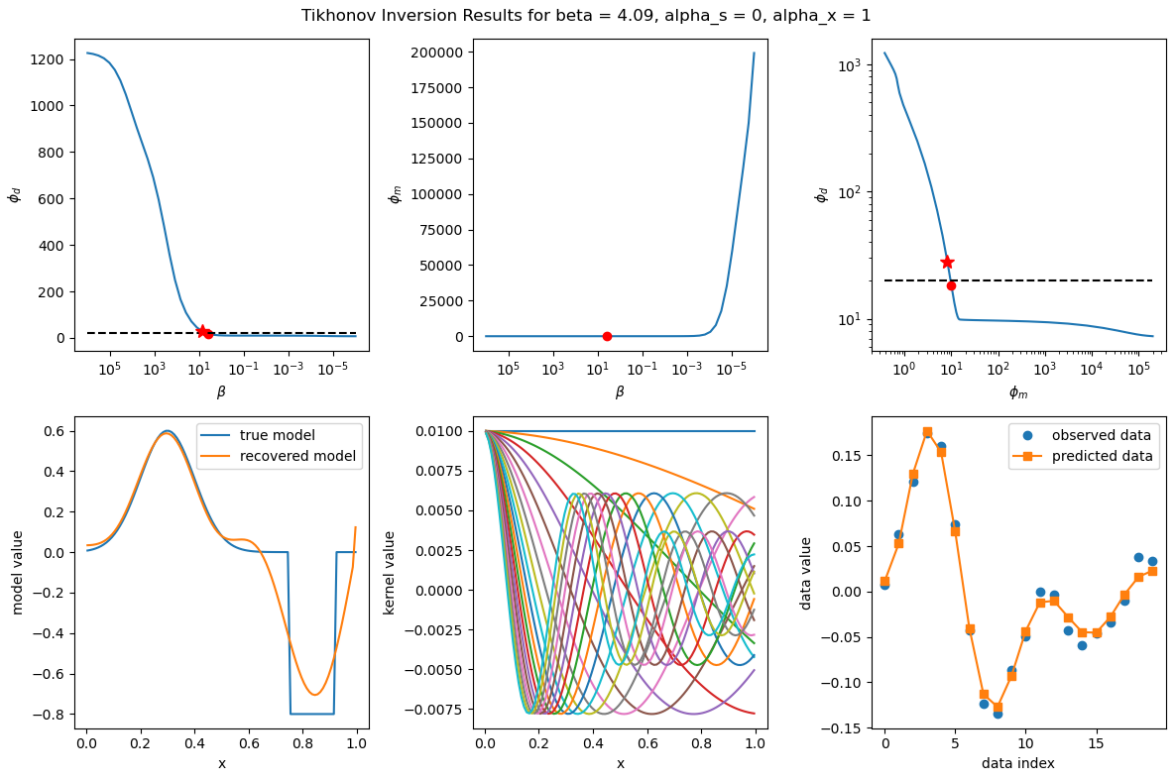

Figure 11: Tikhonov Curves, model, kernels, data for $\alpha_s 0$, $\alpha_x = 1$. In this case we have reached the target misfit and the data fit is acceptable.

3.3 3c

For this section I decided to explore the effects of kernel decay and growth on model fitting. I chose various p values ranging from -0.5 (rapid decay) to 0.1 (growth), and found beta resulting in ideal misfit at each p level. I also set $\alpha_s = \alpha_x = 1$ for this experiment. The resulting kernels are shown in figure 13, and the resulting model fits are shown in figure 14. These plots seem to demonstrate that increasing or decreasing the p value from our starting value of -0.05 makes the fit significantly worse. Models with p too high and low both resulted in oscillatory fits, even at ideal beta.

4 Problem 4: Singular Value Decomposition

4.1 4a

See figure 15.

4.2 4b

See "get_m_svd" function in notebook.

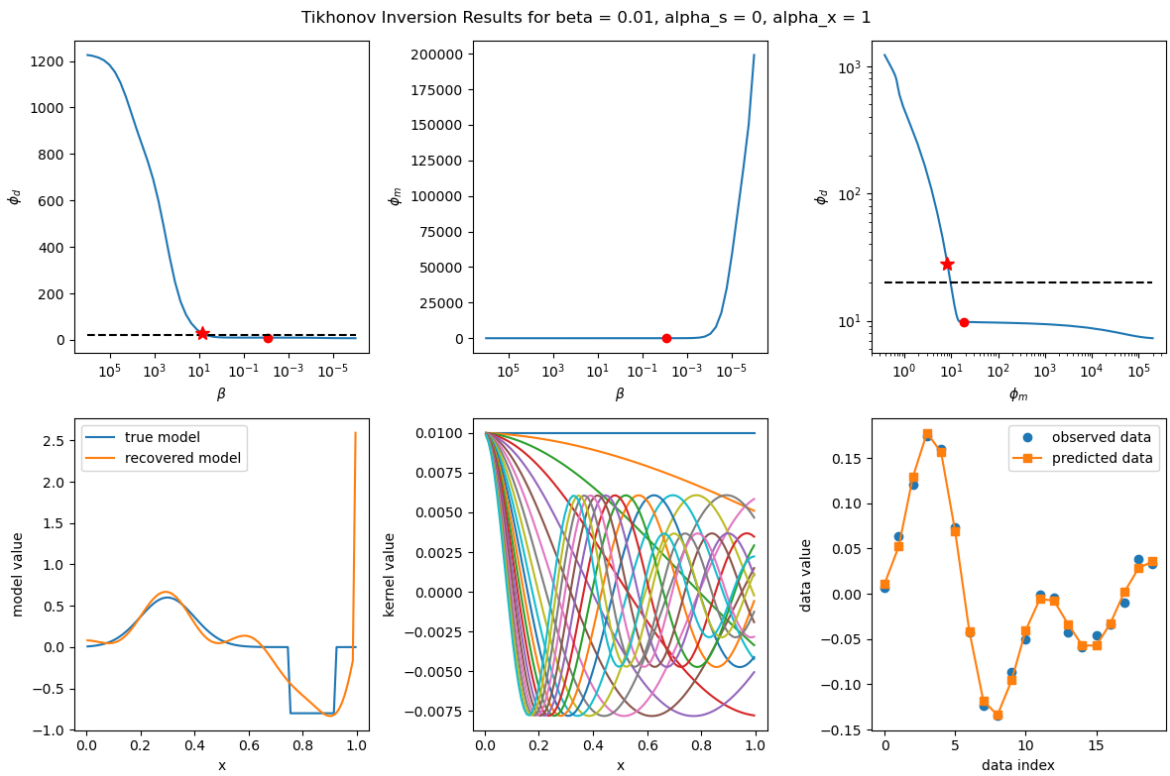

Figure 12: Tikhonov Curves, model, kernels, data for $\alpha_s = 1$, $\alpha_x = 0$. In this case we have overshot the target misfit, and the recovered model is beginning to diverge from the true model.

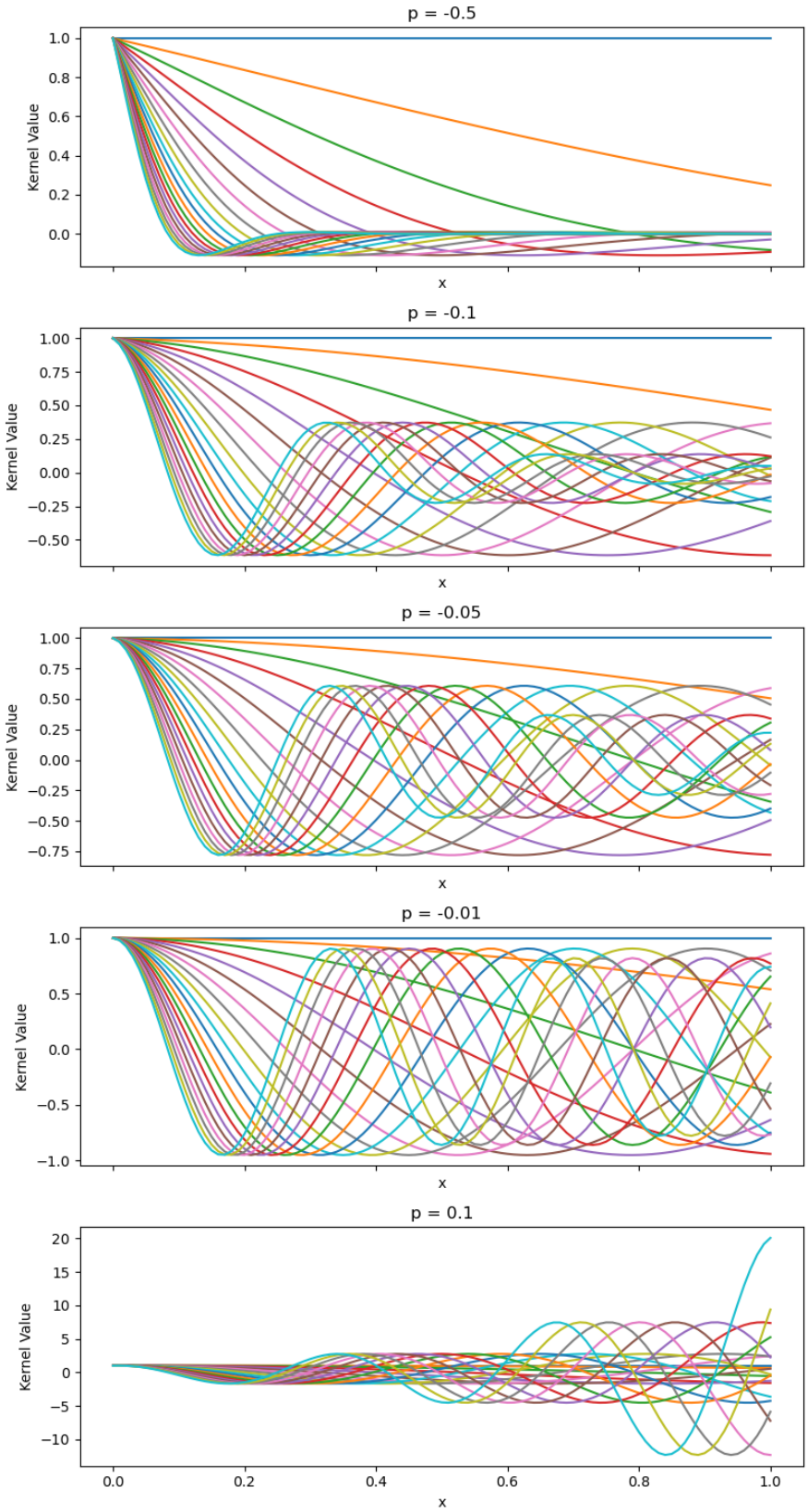

Figure 13: The sets of kernels used to generate models with. p value labelled in each title.

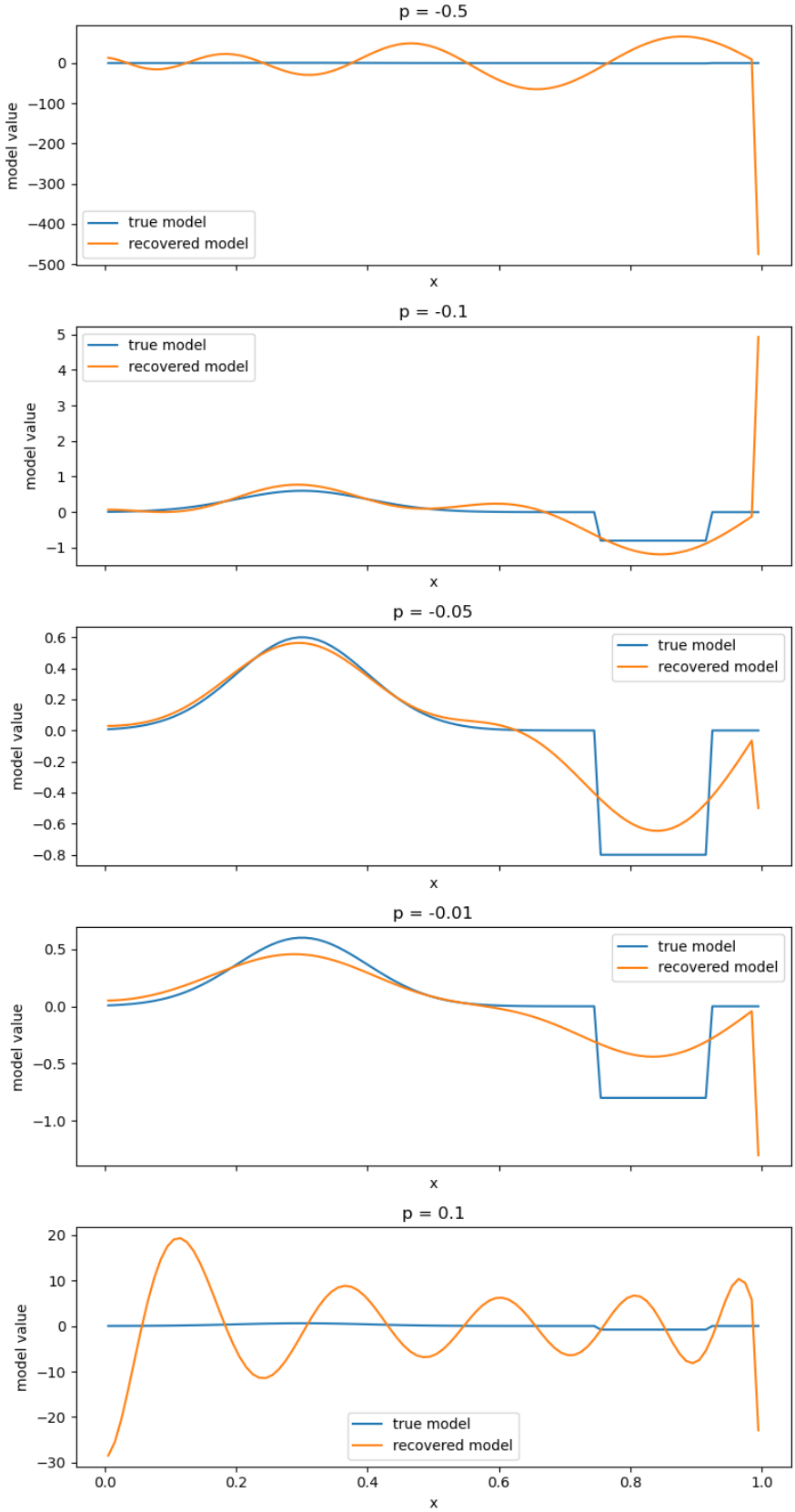

Figure 14: Ideal model fits using various p values. Medium values appear to give the best fits.

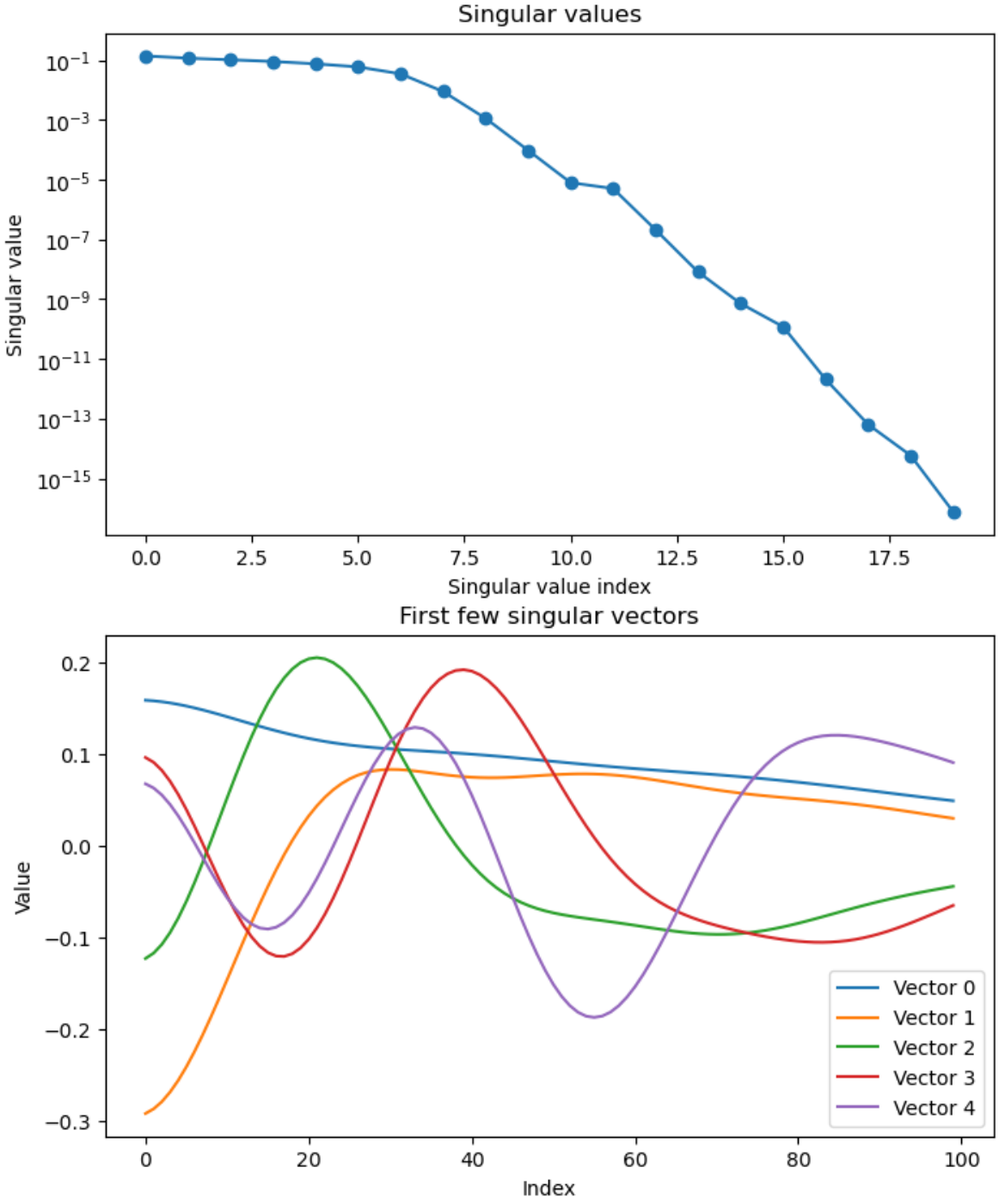

Figure 15: Singular values and the first few singular vectors of the G matrix

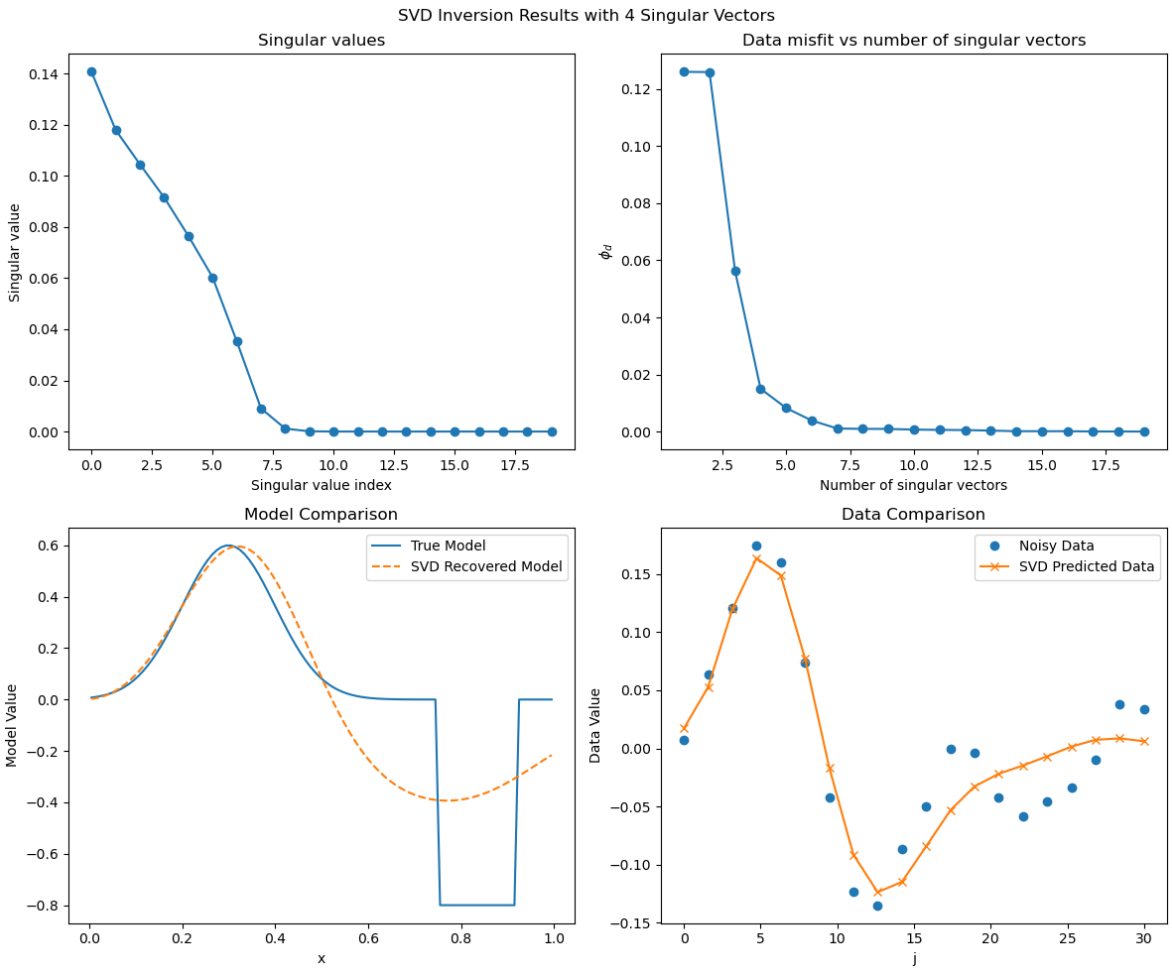

Figure 16: Underfitting the data using the first four singular vectors in the SVD method.

4.3 4c

See figures 16, 17, 18.

4.4 4d

For this section I decided to explore how changing the frequency of oscillations of in the kernels by changing the q values affects the model fit. I chose q values ranging from 0.01 (slow oscillation) to 0.2 (fast oscillation), and tried to find the ideal model fit using SVD for each. See figure 19 for the sets of kernels used, and figure 20 for the resulting fits. As when I varied p, I found that changing the q value from its original value of 0.1 in either direction made the fits worse, no matter how many singular vectors used. I also found that lowering the frequency of the kernels, resulted in less singular vectors being used before the model fit began to diverge.

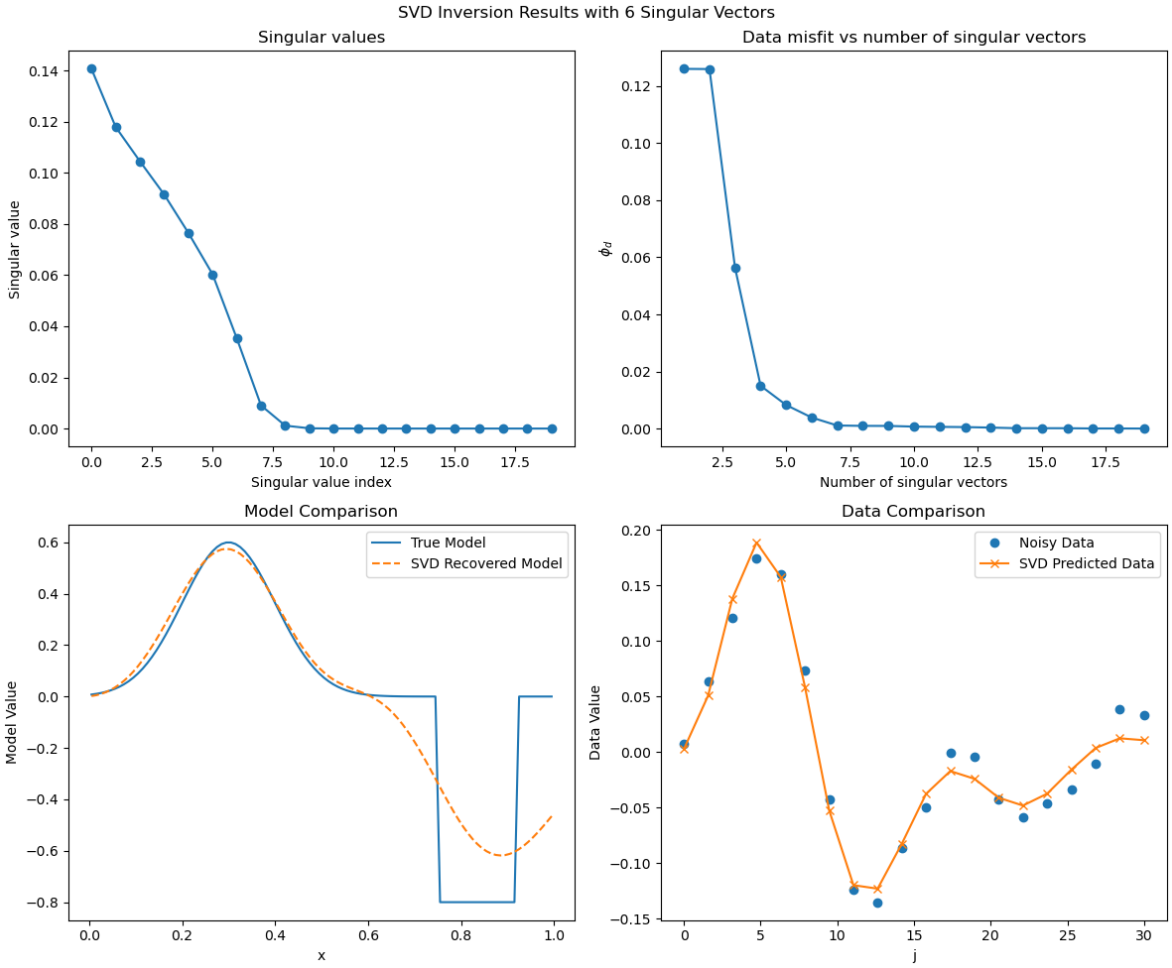

Figure 17: Acceptable data fit using the first six singular vectors in the SVD method.

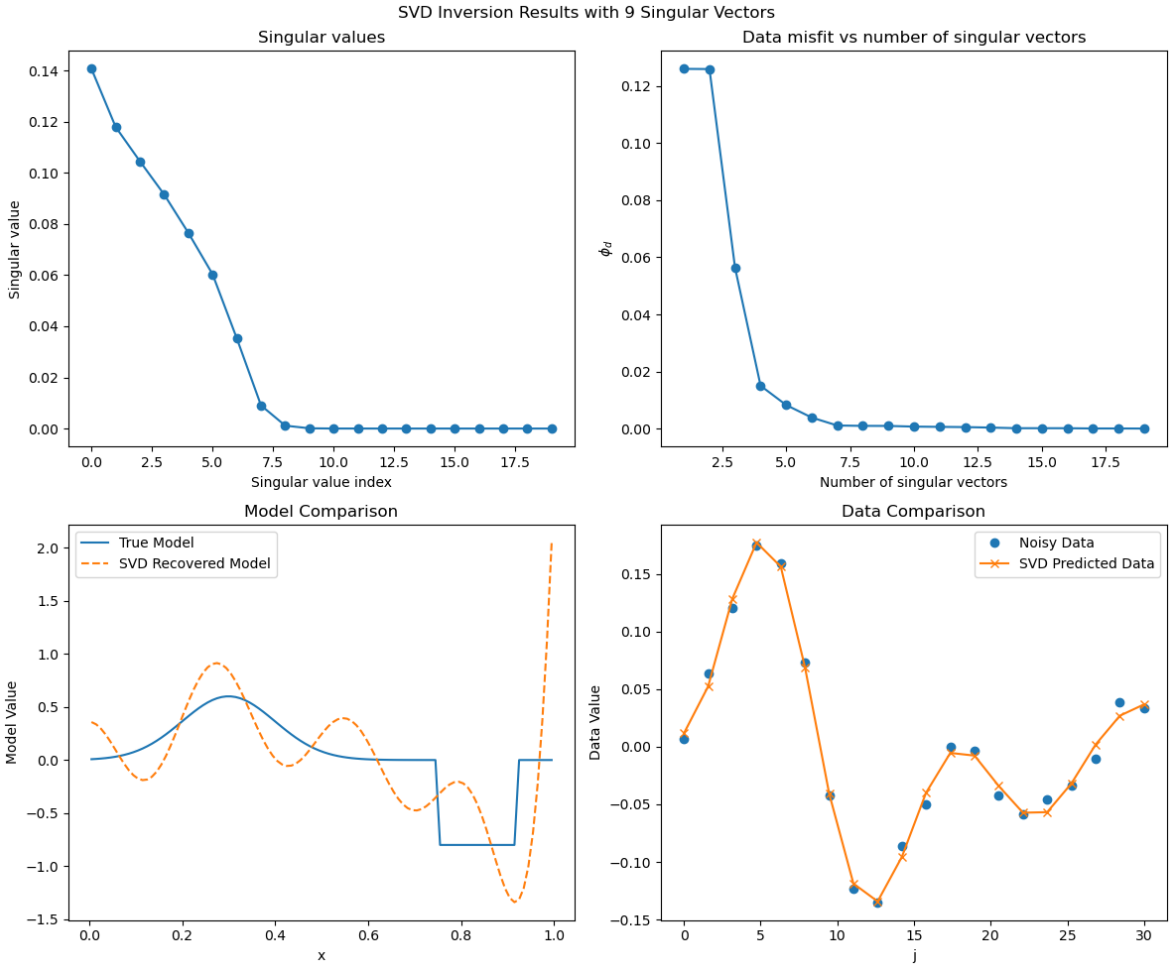

Figure 18: Overfitting the data using the first nine singular vectors in the SVD method.

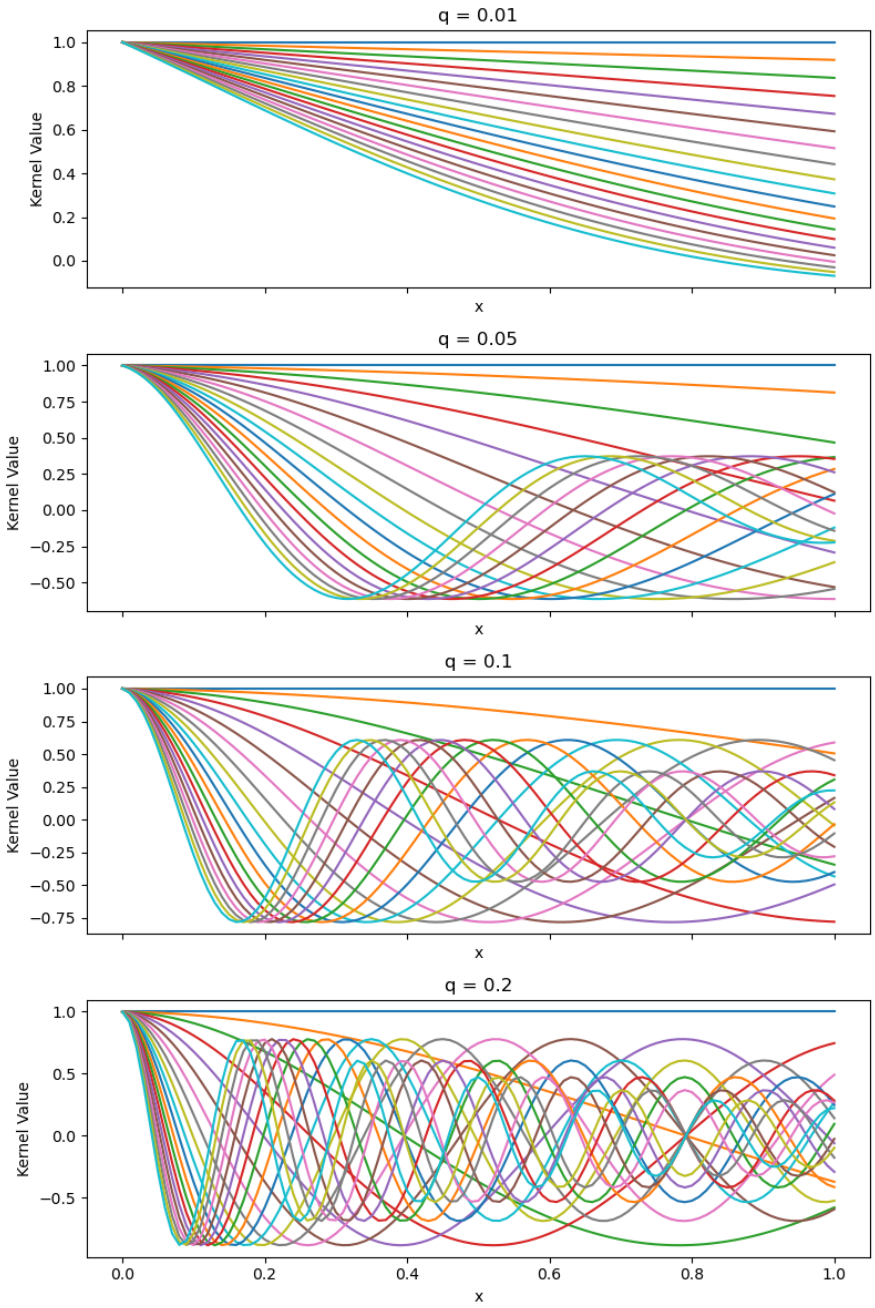

Figure 19: Kernel sets for different q values in the experiment, ranging from slow to fast oscillations.

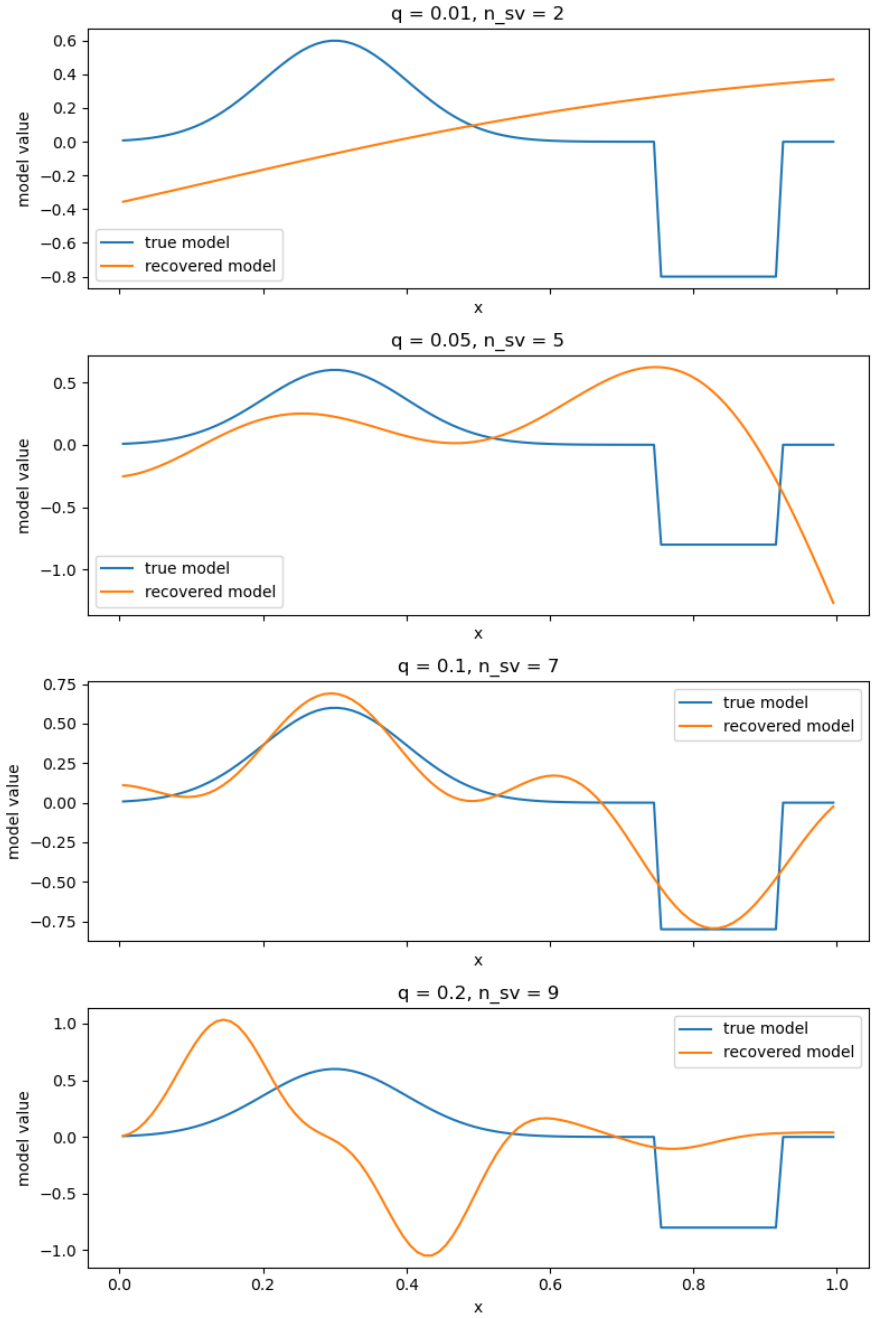

Figure 20: Best model fits for each q value using SVD method. Higher frequencies required more singular vectors to reach best fit.